

RS608 Deep Groove Ball Bearing

Component Specification Sheet ? CBU Gyroscope Project 2026

Dimensional Specifications

Bore diameter (d)	8 mm
Outside diameter (D)	22 mm
Width (B)	7 mm
Fillet radius (r min)	0.3 mm
Mass (approx.)	0.013 kg

Operating Characteristics

Basic dynamic load rating (C)	3.45 kN
Basic static load rating (C0)	1.37 kN
Limiting speed (grease)	32,000 RPM
Limiting speed (oil)	38,000 RPM
Operating temperature range	-20 degC to +100 degC

Construction Details

Type	Deep groove ball bearing
Shield / seal type	ZZ (both sides shielded)
Cage material	Steel
Lubrication	Grease pre-filled
Bore tolerance	h5 (precision)
OD tolerance	h6

Selection Rationale (Project-Specific)

Chosen because	Low friction, high speed rating (>32,000 RPM)
Application speed	approx. 8,000 RPM (well within limit)
Load type	Radial (flywheel weight + centrifugal forces)
Lubrication chosen	Grease (maintenance-free for 24-hour run)
Quantity used	2 (one at each end of the main shaft)

Equivalent Standards / Cross References

ISO designation	608
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ABEC rating	ABEC 3 (standard precision)
Common part numbers	RS608, 608ZZ, 608-2Z
Manufacturer equivalents	SKF 608-2Z, NSK 608ZZ, FAG 608-2RSR

Specification summary compiled for CBU Final Year Project 2026.